# A NODAL CURVE: AN EXPLICIT PROPER HYPERCOVER, ITS DIFFERENTIALS, AND ITS MIXED HODGE STRUCTURE

We compute one rational nodal curve in five steps: *normalization, geometric maps, cohomological pullbacks, spectral sequence, and MHS.* The final two sections give an involution and the positive-genus variant.

## 1. The curve and its normalization

### 1.1. The curve

Let

$$C := \{y^2 z = x^2(x+z)\} \subset \mathbb{P}^2_{\mathbb{C}}, \qquad o = [0:0:1].$$

The affine equation at $o$ is $y^2 = x^2 + x^3$; its quadratic part $(y-x)(y+x)$ has distinct factors, so $o$ is an ordinary node. The curve has no other singularities.

### 1.2. The normalization map

Its normalization is

$$\nu : \mathbb{P}^1 \longrightarrow C, \qquad [u:v] \longmapsto [v(u^2-v^2) : u(u^2-v^2) : v^3].$$

These homogeneous polynomials have no common zero and satisfy the equation of $C$. On $v \neq 0$, with $t = u/v$, this is

$$x = t^2 - 1, \qquad y = t(t^2-1),$$

and $t = y/x$ is the inverse away from the node and infinity. The map is finite and birational, hence is the normalization.

### 1.3. The two branches over the node

The two points over $o$ are

$$a = [1:1], \qquad b = [-1:1], \qquad \nu^{-1}(o) = \{a, b\}.$$

Away from these two points, $\nu$ is an isomorphism onto $C \setminus \{o\}$.

## 2. The hypercover and its geometric maps

### 2.1. The levels $X_p$

Take the reduced Čech nerve of $\nu$:

$$X_p := \Big(\underbrace{\mathbb{P}^1 \times_C \cdots \times_C \mathbb{P}^1}_{p+1 \text{ factors}}\Big)_{\mathrm{red}} \qquad (p \geq 0).$$

Write $D_p \cong \mathbb{P}^1$ for its diagonal component. For each nonconstant word $\epsilon = (\epsilon_0, \ldots, \epsilon_p)$ in $\{a, b\}$, write $P_\epsilon$ for the corresponding reduced point. Then, as schemes,

$$\boxed{X_p = D_p \sqcup \coprod_{\substack{\epsilon \in \{a,b\}^{p+1} \\ \epsilon \text{ nonconstant}}} P_\epsilon.}$$

Indeed, a tuple over a smooth point is diagonal. A tuple over $o$ has entries in $\{a, b\}$; its two constant tuples belong to the diagonal, and all other tuples are isolated reduced points after reduction. The diagonal is closed; the remaining finitely many points are closed and disjoint from it, so the displayed union is a disjoint union of reduced schemes. In particular, every $X_p$ is smooth and projective. The first three levels are

$$\begin{aligned} X_0 &= \mathbb{P}^1, \\ X_1 &= D_1 \sqcup P_{ab} \sqcup P_{ba}, \\ X_2 &= D_2 \sqcup P_{aab} \sqcup P_{aba} \sqcup P_{abb} \sqcup P_{baa} \sqcup P_{bab} \sqcup P_{bba}. \end{aligned}$$

Thus $X_p$ has $2^{p+1} - 1$ connected components. The augmentation $\pi_p : X_p \to C$ is $\nu$ on $D_p$ and sends every isolated point to $o$.

### 2.2. Faces: delete one coordinate

For $p \geq 1$ and $0 \leq i \leq p$, the face map is

$$d_i^{(p)} : X_p \longrightarrow X_{p-1}, \qquad (x_0, \dots, x_p) \longmapsto (x_0, \dots, \widehat{x_i}, \dots, x_p).$$

### 2.3. Degeneracies: repeat one coordinate

For $p \geq 0$ and $0 \leq i \leq p$, the degeneracy map is

$$s_i^{(p)} : X_p \longrightarrow X_{p+1}, \qquad (x_0, \dots, x_p) \longmapsto (x_0, \dots, x_i, x_i, \dots, x_p).$$

They descend from the Čech nerve to its reductions. On the diagonal components all these maps are the identity of $\mathbb{P}^1$. On a point component they delete or repeat the indicated letter. If deletion produces a constant word $aa \cdots a$ or $bb \cdots b$, the image is the point $a$ or $b$ on $D_{p-1}$, not an additional point component.

### 2.4. Images of points: $X_1 \to X_0$

In degree one,

$$d_0(P_{ab}) = b, \quad d_1(P_{ab}) = a, \qquad d_0(P_{ba}) = a, \quad d_1(P_{ba}) = b,$$

whereas $s_0^{(0)} : \mathbb{P}^1 \to X_1$ includes $D_1$.

### 2.5. Images of points: $X_2 \to X_1$

Each row labels a point $P_\epsilon \in X_2$. The entry in column $d_i^{(2)}$ is its image in $X_1$ after deleting coordinate $i$. For example,

$$(a, a, b) \overset{d_0}{\longmapsto} (a, b), \qquad (a, a, b) \overset{d_1}{\longmapsto} (a, b), \qquad (a, a, b) \overset{d_2}{\longmapsto} (a, a).$$

Here $(a, a)$ means $a \in D_1 \cong \mathbb{P}^1$; similarly, $(b, b)$ means $b \in D_1$. This is a table of geometric images.

| | $d_0^{(2)}$ | $d_1^{(2)}$ | $d_2^{(2)}$ |
|---|---|---|---|
| $P_{aab}$ | $P_{ab}$ | $P_{ab}$ | $a \in D_1$ |
| $P_{aba}$ | $P_{ba}$ | $a \in D_1$ | $P_{ab}$ |
| $P_{abb}$ | $b \in D_1$ | $P_{ab}$ | $P_{ab}$ |
| $P_{baa}$ | $a \in D_1$ | $P_{ba}$ | $P_{ba}$ |
| $P_{bab}$ | $P_{ab}$ | $b \in D_1$ | $P_{ba}$ |
| $P_{bba}$ | $P_{ba}$ | $P_{ba}$ | $b \in D_1$ |

### 2.6. Images of points under $s_i^{(1)} : X_1 \to X_2$

Each entry is the image of the row point after repeating coordinate $i$:

| | $s_0^{(1)}$ | $s_1^{(1)}$ |
|---|---|---|
| $P_{ab}$ | $P_{aab}$ | $P_{abb}$ |
| $P_{ba}$ | $P_{bba}$ | $P_{baa}$ |

Both degeneracies send $D_1$ identically to $D_2$.

### 2.7. The simplicial identities

Deleting and repeating coordinates directly give the simplicial identities (superscripts are suppressed):

$$d_i d_j = d_{j-1} d_i \quad (i < j), \qquad s_i s_j = s_{j+1} s_i \quad (i \leq j),$$
$$d_i s_j = \begin{cases} s_{j-1} d_i, & i < j, \\ \mathrm{id}, & i = j \text{ or } i = j+1, \\ s_j d_{i-1}, & i > j+1. \end{cases}$$

The augmentations satisfy $\pi_{p-1} d_i^{(p)} = \pi_p$ and $\pi_{p+1} s_i^{(p)} = \pi_p$.

### 2.8. Proper hypercover and descent

Let $M_p = (\operatorname{cosk}_{p-1}^C X_{\leq p-1})_p$ denote the matching scheme, with $M_0 = C$. Thus $M_1 = X_0 \times_C X_0$, and for $p \geq 2$ its geometric points are compatible collections of the $p+1$ faces of a $p$-simplex. All the maps $X_p \to M_p$ are proper. They are surjective on geometric points: reduction does not change geometric points, and a compatible collection of faces in a Čech nerve uniquely specifies its tuple of vertices over a common point of $C$. For $p = 1$ this just specifies two vertices over the same point. For $p = 0$ it is the surjectivity of $\nu$. This verifies the proper hypercover condition [D74, (5.3.8)]. Consequently proper descent [D74, (8.2.1.2)] gives

$$\pi_\bullet^* : H^n(C, \mathbb{Q}) \xrightarrow{\sim} H^n(X_\bullet, \mathbb{Q}).$$

## 3. The cochain complex and the first page

### 3.1. Singular cochains and their differential

Put

$$K^{p,q} = C^q_{\mathrm{sing}}(X_p^{\mathrm{an}}, \mathbb{Q}), \qquad K^{p,q} = 0 \text{ if } p < 0 \text{ or } q < 0.$$

For a singular cochain $\alpha$, the usual differential is

$$(d_{\mathrm{sing}}\alpha)(\tau) = \sum_{j=0}^{q+1} (-1)^j \alpha(\tau \circ \partial_j), \qquad \tau : \Delta^{q+1} \to X_p^{\mathrm{an}}.$$

Here $\partial_j$ is a face inclusion of the topological simplex; it is distinct from the algebraic face map $d_i^{(p)}$.

### 3.2. Horizontal, vertical, and total differentials

Define

$$h^{p,q} = \sum_{i=0}^{p+1} (-1)^i (d_i^{(p+1)})^*, \quad v^{p,q} = (-1)^p d_{\mathrm{sing}},$$
$$\operatorname{Tot}^n K = \bigoplus_{p+q=n} K^{p,q}, \qquad D = h + v.$$

Thus $h$ increases $p$ by one and $v$ increases $q$ by one. Naturality and the simplicial identities give $h^2 = v^2 = hv + vh = 0$, hence $D^2 = 0$. The degeneracy pullbacks

$$(s_i^{(p-1)})^* : K^{p,q} \longrightarrow K^{p-1,q}$$

are not extra summands in $D$; they will remove the redundant cosimplicial directions in the row computations below.

### 3.3. Notation: the Tate Hodge structure $\mathbb{Q}(i)$

For an integer $i$, $\mathbb{Q}(i)$ denotes a one-dimensional rational pure Hodge structure of weight $-2i$ and Hodge type $(-i,-i)$. Writing $V=\mathbb{Q}(i)$, its filtrations are

$$W_wV=\begin{cases}0, & w<-2i,\\ V, & w\geq -2i,\end{cases}\qquad F^\ell V_{\mathbb{C}}=\begin{cases}V_{\mathbb{C}}, & \ell\leq -i,\\ 0, & \ell>-i.\end{cases}$$

Its underlying rational vector space is isomorphic to $\mathbb{Q}$; the standard Tate convention realizes it as $(2\pi\sqrt{-1})^i\mathbb{Q}$ inside $\mathbb{C}$. The integer in parentheses records the Tate twist, not the cohomological degree. In this example we use

| Notation | Weight | Hodge type | Example |
|---|---|---|---|
| $\mathbb{Q}(0)$ | 0 | $(0,0)$ | $H^0(\mathbb{P}^1,\mathbb{Q})$ |
| $\mathbb{Q}(-1)$ | 2 | $(1,1)$ | $H^2(\mathbb{P}^1,\mathbb{Q})$ |

The superscript in $\mathbb{Q}(0)^m$ denotes a direct sum of $m$ copies, so its rational dimension is $m$.

### 3.4. The first page $E_1$

The column filtration yields

$$E_1^{p,q}=H^q(X_p,\mathbb{Q}),\qquad d_1^{p,q}=\delta^{p,q}:=\sum_{i=0}^{p+1}(-1)^i(d_i^{(p+1)})^*,\qquad E_1^{p,q}\Longrightarrow H^{p+q}(C,\mathbb{Q}).$$

Here $\delta$ is induced by $h$, after taking cohomology in the vertical direction. In our example,

$$H^q(X_p,\mathbb{Q})=\begin{cases}\mathbb{Q}^{2^{p+1}-1}, & q=0,\\ \mathbb{Q}(-1), & q=2,\\ 0, & q\neq 0,2.\end{cases}$$

## 4. Pullbacks on cohomology

### 4.1. The pullback rule on $H^0$

The geometric map $d_i^{(p+1)}:X_{p+1}\to X_p$ induces $(d_i^{(p+1)})^*:H^0(X_p,\mathbb{Q})\to H^0(X_{p+1},\mathbb{Q})$ in the opposite direction. A vector $c\in H^0(X_p,\mathbb{Q})$ is a locally constant function: it has a value $c_D$ on $D_p$ and a value $c_\epsilon$ on each isolated component. For any component $A$ of $X_{p+1}$,

$$((d_i^{(p+1)})^*c)(A)=c(d_i(A)),$$

where a point of $D_p$ has value $c_D$. Likewise, for a component $A$ of $X_{p-1}$,

$$((s_i^{(p-1)})^*c)(A)=c(s_i(A)).$$

For instance, $d_0(P_{aab})=P_{ab}$ gives $(d_0^*c)(P_{aab})=c(P_{ab})$.

### 4.2. The pullback rule on $H^2$

On $H^2$, every face pullback and every degeneracy pullback is the identity of $\mathbb{Q}(-1)$, because its restriction between the diagonal components is the identity. Point components have no $H^2$. There are no other nonzero cohomology groups.

**4.3. The three maps $d_i^* : H^0(X_1) \to H^0(X_2)$**

Order the components of $X_1$ as $(D_1, P_{ab}, P_{ba})$ and write $c = (c_D, A, B)$. Order those of $X_2$ as in Section 2. The face table gives the three maps $H^0(X_1) \to H^0(X_2)$:

$$\begin{aligned} d_0^* c &= (c_D, A, B, c_D, c_D, A, B), \\ d_1^* c &= (c_D, A, c_D, A, B, c_D, B), \\ d_2^* c &= (c_D, c_D, A, A, B, B, c_D). \end{aligned}$$

**4.4. The alternating sum $\delta = d_0^* - d_1^* + d_2^* - \cdots$**

Both maps $d_0^*, d_1^* : H^0(X_0) \to H^0(X_1)$ send $c_D$ to $(c_D, c_D, c_D)$. Therefore

$$\delta^{0,0} = 0, \qquad \delta^{1,0}(c_D, A, B) = (c_D, c_D, A + B - c_D, c_D, c_D, A + B - c_D, c_D).$$

**4.5. The kernel modulo image**

Consequently,

$$E_2^{1,0} = \ker \delta^{1,0} / \operatorname{im} \delta^{0,0} = \{(0, t, -t) : t \in \mathbb{Q}\} \cong \mathbb{Q}.$$

**4.6. The degeneracy pullbacks $s_i^*$**

For $z = (z_D, z_{aab}, z_{aba}, z_{abb}, z_{baa}, z_{bab}, z_{bba})$, the pullbacks are

$$s_0^*(c_D, A, B) = c_D, \qquad s_0^* z = (z_D, z_{aab}, z_{bba}), \qquad s_1^* z = (z_D, z_{abb}, z_{baa}).$$

## 5. The second page and total cohomology

**5.1. The normalized row complex**

For a fixed $q$ write $A_q^p = H^q(X_p, \mathbb{Q})$. Its normalized cosimplicial cochain complex is

$$N^0 A_q = A_q^0, \qquad N^p A_q = \bigcap_{i=0}^{p-1} \ker\Big(s_i^* : A_q^p \to A_q^{p-1}\Big) \quad (p \geq 1),$$

with the restriction of $\delta$ as differential. The standard cosimplicial normalization theorem says that the inclusion $NA_q \hookrightarrow (A_q^\bullet, \delta)$ is a quasi-isomorphism [Stacks, Sections 14.24–14.25]. Thus it computes the same $E_2^{p,q}$. We are simplifying the complex on the $E_1$ row; we are not asserting that $N^p A_q = E_1^{p,q}$.

**5.2. The row $q = 0$**

For $q = 0$, belonging to $N^p A_0$ means vanishing on every component in the image of a degeneracy. When $p \geq 1$ the whole diagonal is degenerate. A nonconstant word is degenerate exactly when two adjacent letters are equal. The only nondegenerate words of length $p+1$ are the two alternating words, starting with $a$ and $b$, respectively. Therefore

$$N^0 A_0 = \mathbb{Q}, \qquad N^p A_0 = \mathbb{Q}^2 \quad (p \geq 1).$$

Use coordinates $(A, B)$ given by these two alternating words. In an alternating word, deleting an interior letter makes two adjacent letters equal. Only the two endpoint deletions contribute to the normalized differential. Hence, for $p \geq 1$,

$$\Delta_p(A, B) = \Big(B + (-1)^{p+1} A,\ A + (-1)^{p+1} B\Big), \qquad \Delta_0 = 0.$$

Equivalently the row is

$$\mathbb{Q} \xrightarrow{0} \mathbb{Q}^2 \xrightarrow{S} \mathbb{Q}^2 \xrightarrow{A} \mathbb{Q}^2 \xrightarrow{S} \mathbb{Q}^2 \xrightarrow{A} \cdots, \qquad S = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad A = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}.$$

One checks directly that

$$\ker S = \operatorname{im} A = \mathbb{Q}(1,-1), \qquad \ker A = \operatorname{im} S = \mathbb{Q}(1,1).$$

Thus

$$E_2^{0,0} = \mathbb{Q}, \quad E_2^{1,0} = \mathbb{Q}, \quad E_2^{p,0} = 0 \quad (p \geq 2).$$

### 5.3. The row $q = 2$

For $q = 2$, all degeneracy pullbacks are identities, so $N^0 A_2 = \mathbb{Q}(-1)$ and $N^p A_2 = 0$ for $p \geq 1$. As a check, the unnormalized differential is

$$\delta^{p,2} = \left( \sum_{i=0}^{p+1} (-1)^i \right) \mathrm{id} = \begin{cases} 0, & p \text{ even}, \\ \mathrm{id}, & p \text{ odd}, \end{cases}$$

so this row is $\mathbb{Q}(-1) \xrightarrow{0} \mathbb{Q}(-1) \xrightarrow{\mathrm{id}} \mathbb{Q}(-1) \xrightarrow{0} \cdots$. It follows that $E_2^{0,2} = \mathbb{Q}(-1)$ and $E_2^{p,2} = 0$ for $p > 0$.

### 5.4. The $E_2$ table

The complete nonzero part of $E_2$ is

| $q\backslash p$ | 0 | 1 | $p \geq 2$ |
|---|---|---|---|
| 2 | $\mathbb{Q}(-1)$ | 0 | 0 |
| 1 | 0 | 0 | 0 |
| 0 | $\mathbb{Q}(0)$ | $\mathbb{Q}(0)$ | 0 |

### 5.5. Higher differentials and convergence

For $r \geq 2$, the differentials have bidegree $(r, 1-r)$. There is no nonzero source and target pair in this table: in particular $d_2^{0,2}$ has zero target $E_2^{2,1}$, and $d_3^{0,2}$ has zero target $E_3^{3,0}$. Thus every higher differential vanishes and $E_2 = E_\infty$. Finite column filtrations in each total degree give convergence, so the underlying vector spaces are

$$H^0(C,\mathbb{Q}) = \mathbb{Q}, \qquad H^1(C,\mathbb{Q}) = \mathbb{Q}, \qquad H^2(C,\mathbb{Q}) = \mathbb{Q}, \qquad H^n(C,\mathbb{Q}) = 0 \ (n > 2).$$

## 6. The mixed Hodge structure

### 6.1. Constructing the weight filtration $W$

On the rational total complex set

$$L^p \operatorname{Tot}^m K := \bigoplus_{i \geq p} K^{i,m-i}, \qquad W_w H^n(X_\bullet, \mathbb{Q}) := \operatorname{im}\Big( H^n(L^{n-w} \operatorname{Tot} K) \to H^n(\operatorname{Tot} K) \Big).$$

### 6.2. Constructing the Hodge filtration $F$

Let $\mathcal{A}^{a,b}(X_p)$ be smooth complex differential forms of type $(a,b)$ and put

$$B^m = \bigoplus_{p+a+b=m} \mathcal{A}^{a,b}(X_p), \qquad D_B = h + (-1)^p d_{\mathrm{dR}}, \qquad F^\ell B^m = \bigoplus_{\substack{p+a+b=m \\ a \geq \ell}} \mathcal{A}^{a,b}(X_p).$$

Through the natural de Rham comparison $H^n(B) \cong H^n(X_\bullet, \mathbb{C})$, define

$$F^\ell H^n(X_\bullet, \mathbb{C}) = \operatorname{im}\Big( H^n(F^\ell B) \longrightarrow H^n(B) \Big).$$

### 6.3. Reading the MHS from $E_2$

Deligne's construction says that these are the MHS filtrations and that

$$\mathrm{Gr}^W_w\, H^n(X_\bullet, \mathbb{Q}) \cong E_2^{n-w,w}$$

as pure Hodge structures of weight $w$ [D74, (8.1.20)]. Transport the filtrations to $H^n(C, \mathbb{Q})$ by $(\pi_\bullet^*)^{-1}$. Independence of the hypercover is Deligne's theorem [D74, (8.2.2)]. Our explicit $E_2$ table gives

$$\boxed{H^0(C, \mathbb{Q}) \cong \mathbb{Q}(0), \qquad H^1(C, \mathbb{Q}) \cong \mathbb{Q}(0), \qquad H^2(C, \mathbb{Q}) \cong \mathbb{Q}(-1).}$$

These are isomorphisms of Hodge structures. In particular, $H^1$ has weight 0, even though its cohomological degree is 1.

### 6.4. The $E_1$ diagram: cohomology of the individual levels

Columns are simplicial degrees $p$; rows are cohomological degrees $q$. The schemes at the foot of the columns are

$$X_0 = \mathbb{P}^1, \quad X_1 = \mathbb{P}^1 \sqcup \{P_{ab}, P_{ba}\}, \quad X_2 = \mathbb{P}^1 \sqcup 6 \text{ points}, \quad X_3 = \mathbb{P}^1 \sqcup 14 \text{ points}.$$

Each entry is $E_1^{p,q} = H^q(X_p, \mathbb{Q})$. All displayed arrows are $d_1^{p,q} = \delta^{p,q}$, the alternating sums of face pullbacks.

$$\begin{array}{c|ccccccccc}
q & & & & & & & & & \\
2 & \mathbb{Q}(-1) & \xrightarrow{0} & \mathbb{Q}(-1) & \xrightarrow{\mathrm{id}} & \mathbb{Q}(-1) & \xrightarrow{0} & \mathbb{Q}(-1) & \cdots & \\
1 & 0 & \xrightarrow{0} & 0 & \xrightarrow{0} & 0 & \xrightarrow{0} & 0 & \cdots & \\
0 & \mathbb{Q}(0) & \xrightarrow{\delta^{0,0}=0} & \mathbb{Q}(0)^3 & \xrightarrow{\delta^{1,0}} & \mathbb{Q}(0)^7 & \xrightarrow{\delta^{2,0}} & \mathbb{Q}(0)^{15} & \cdots & \\
\hline
 & 0 & & 1 & & 2 & & 3 & & p \\
 & X_0 & & X_1 & & X_2 & & X_3 & &
\end{array}$$

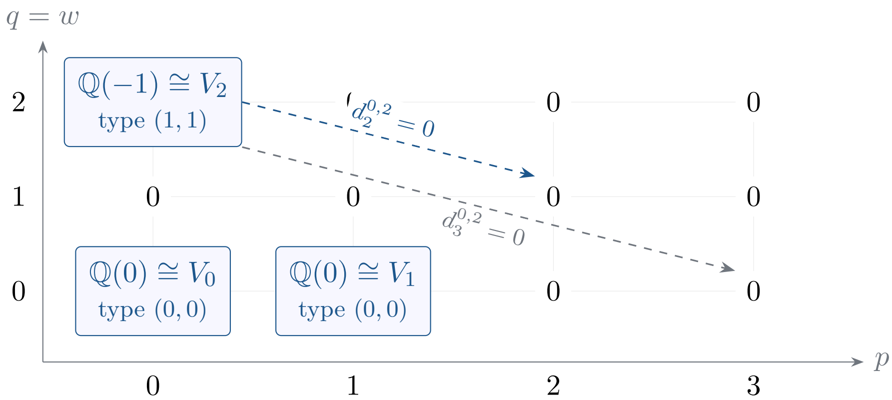
The bottom row records the node. In the coordinates $(D_1, P_{ab}, P_{ba})$ of $H^0(X_1) = \mathbb{Q}^3$,

$$\delta^{1,0}(c, A, B) = (c, c, A + B - c, c, c, A + B - c, c),$$

so the incoming image is zero and the outgoing kernel is $\mathbb{Q}(0, 1, -1)$. This produces $E_2^{1,0} = \mathbb{Q}(0)$. All entries in row $q$ are pure of weight $q$.

### 6.5. The $E_2 = E_\infty$ diagram: the surviving MHS pieces

Taking kernel modulo incoming image in each row gives the diagram below. Blue boxes are the surviving terms; all other terms are zero. Write $V_n = H^n(X_\bullet, \mathbb{Q})$. Each $V_n$ here has only one nonzero weight-graded piece, shown on the diagonal $p + q = n$.




The dashed arrows illustrate the bidegree $d_r : (p, q) \to (p + r, q - r + 1)$. The $d_3$ arrow is drawn on the same lattice after $E_3 = E_2$. Both targets are zero; every remaining higher differential also has zero source or target. Thus $E_2 = E_\infty$.

### 6.6. How to read $L$, $W$, and $F$ from the diagram

(1) **Column filtration $L$.** Fix a total-degree diagonal $p + q = n$. The pieces belonging to $L^r V_n$ have $p \geq r$:
$$\mathrm{Gr}_L^p V_n \cong E_\infty^{p,n-p}.$$
In particular $L^1 V_1 = V_1$, whereas $L^1 V_0 = L^1 V_2 = 0$.

(2) **Weight filtration $W$.** On that same diagonal, $W_w V_n = L^{n-w} V_n$ consists of the pieces in rows $q \leq w$. Hence $V_0, V_1$ have weight 0, and $V_2$ has weight 2.

(3) **Hodge filtration $F$.** Read the type $(a, b)$ inside each blue box: $F^\ell$ retains types with $a \geq \ell$. Thus $F^1 V_{0,\mathbb{C}} = F^1 V_{1,\mathbb{C}} = 0$ and $F^1 V_{2,\mathbb{C}} = V_{2,\mathbb{C}}$, $F^2 V_{2,\mathbb{C}} = 0$.

Each surviving group is one-dimensional, and $V_n = 0$ for $n > 2$. In particular the box at $(1, 0)$ comes from $H^0(X_1)$, although it contributes to $H^1(X_\bullet)$; $H^1(X_1)$ itself is zero.

### 6.7. A generator of $H^1$

The generator of $E_2^{1,0}$ is the locally constant function
$$\eta \in K^{1,0}, \qquad \eta|_{D_1} = 0, \quad \eta(P_{ab}) = 1, \quad \eta(P_{ba}) = -1.$$
It satisfies $v\eta = 0$ and $h\eta = 0$, by the face table, so it is a closed total cochain of degree one. The same locally constant function is a total differential form in simplicial degree 1 and form type $(0, 0)$, representing the corresponding complex class. Since $\eta \in L^1 \operatorname{Tot}^1 K$, it represents a class of weight 0. Since its form type is $(0, 0)$, it lies in $F^0$. The computed $E_2$ table shows that it is nonzero and spans $H^1$; there are no type $(1, -1)$ classes, so $F^1 H^1 = 0$.

### 6.8. The graded comparison on this generator

In this example the comparison from `MHS_EQUIV` is simply
$$\alpha_{1,0} : \mathrm{Gr}_L^1 H^1(X_\bullet, \mathbb{Q}) \xrightarrow{\sim} \ker \delta^{1,0}, \qquad [\eta]_D \longmapsto (0, 1, -1).$$
Here $\mathrm{Gr}_L^1 H^1 = H^1$, since $L^1 H^1 = H^1$ and $L^2 H^1 = 0$.

### 6.9. A generator of $H^2$

A generator of $H^2$ is represented in the de Rham total complex by a closed $(1,1)$-form $\omega$ on $X_0 = \mathbb{P}^1$ with $\int_{\mathbb{P}^1} \omega = 1$. Its two pullbacks to $D_1$ agree, and its pullbacks to the point components vanish, so $h\omega = 0$. This class has weight 2 and type $(1,1)$. Topologically, $C^{\mathrm{an}}$ is a sphere with $a$ and $b$ identified, homotopy equivalent to $S^2 \vee S^1$. The extra $S^1$ accounts for the weight-zero $H^1$ class above.

## 7. An explicit equivariant example

### 7.1. The involution and its lift

The involution
$$f : C \to C, \qquad [x : y : z] \longmapsto [x : -y : z]$$
lifts to $\widetilde{f} : [u : v] \mapsto [-u : v]$ on $\mathbb{P}^1$ and interchanges $a$ and $b$. It induces a simplicial automorphism
$$f_p(x_0, \dots, x_p) = (\widetilde{f}(x_0), \dots, \widetilde{f}(x_p))$$
which commutes with all coordinate deletions and repetitions.

### 7.2. The induced action on the row complex

On $H^0(X_p)$ its pullback fixes the diagonal coordinate and interchanges each word with the word obtained by swapping $a, b$. On every normalized group $N^p A_0 = \mathbb{Q}^2$ for $p \geq 1$, it is
$$J = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad JS = SJ, \quad JA = AJ.$$
In particular $J(1, -1) = -(1, -1)$, so $f_\bullet^*[\eta] = -[\eta]$.

### 7.3. The action on cohomology

On $H^2(\mathbb{P}^1)$ the pullback is the identity. Writing $\Phi = f^*$, the complete answer is
$$\Phi|_{H^0(C)} = 1, \qquad \Phi|_{H^1(C)} = -1, \qquad \Phi|_{H^2(C)} = 1.$$
For instance the equivariance of $\alpha_{1,0}$ reads $\alpha_{1,0}(-[\eta]) = -(0, 1, -1) = J\alpha_{1,0}([\eta])$.

## 8. A normalization of positive genus

### 8.1. The same construction

Let $C'$ be an irreducible projective curve with exactly one ordinary node, whose normalization $\widetilde{C}'$ has genus $g$, and let $a, b$ be the two points above the node. The same reduced Čech construction has $X'_p = \widetilde{C}' \sqcup (2^{p+1} - 2)$ points. The $q = 0$ calculation is unchanged. On $H^1(\widetilde{C}')$ and $H^2(\widetilde{C}')$ all faces and degeneracies restrict to the identity on the diagonal, so their normalized rows are supported only at $p = 0$.

### 8.2. The two weights of $H^1$

Consequently there is an exact sequence of MHS
$$0 \longrightarrow \mathbb{Q}(0) \longrightarrow H^1(C', \mathbb{Q}) \xrightarrow{\nu^*} H^1(\widetilde{C}', \mathbb{Q}) \longrightarrow 0,$$
with
$$W_{-1}H^1 = 0, \qquad W_0 H^1 \cong \mathbb{Q}(0), \qquad W_1 H^1 = H^1.$$
In particular $\dim H^1(C', \mathbb{Q}) = 2g + 1$ and $\dim F^1 H^1(C', \mathbb{C}) = g$.

### 8.3. The extension is additional data

These graded pieces do not specify a splitting of the extension as MHS. In the rational example $g = 0$, the weight-one quotient is zero; hence its $H^1$ is already pure of weight zero. A singular curve need not have a genuinely mixed $H^1$, whereas for $g > 0$ both weights 0 and 1 occur.

## References

[D74] P. Deligne, *Théorie de Hodge, III*, Publ. Math. Inst. Hautes Études Sci. **44** (1974), 5–77. (5.3.8), (8.1.19)–(8.1.20), (8.2.1.2)–(8.2.2).

[Stacks] The Stacks Project Authors, *The Stacks Project*. Section 14.24, Dold–Kan; Section 14.25, Dold–Kan for cosimplicial objects; Section 12.24, spectral sequences of filtered complexes.